Practice Solutions for Midterm 1

1. This problem is all about Lorentz transformations:

   (a) For $F \to F'$, $v = 3c/5$ and $\gamma = 5/4$. The Lorentz transformations are thus:

$$
\begin{aligned}
t' &= \frac{5}{4}(t - \frac{3}{5c}x) \\
x' &= \frac{5}{4}(x - \frac{3c}{5}t) \\
y' &= y \\
z' &= z
\end{aligned}
$$

   (b) These can be read off from the Lorentz transformation: $(-1/4, 7/4, 3, 4)$.

   (c) We need to relativistically add $4c/5$ and $-3c/5$; the minus sign because the rocket is moving in the same direction as R' as seen in frame R. Let u be the speed of the rocket and v the speed between R and R':

$$u^{x'} = \frac{u+v}{1+uv/c^2} = \frac{c/5}{1-12/25} = \frac{5c}{13}\ .$$

   (d) The rocket's length in this frame is given by a length contraction of its rest frame length using the Lorentz factor corresponding to its speed in R':

$$\gamma = \frac{1}{\sqrt{1-(5/13)^2}} = \frac{13}{12}\ ,$$

   so

$$L = \frac{12L_0}{13}\ .$$

2. (a) No: the volume of a meter stick changes when its length is measured to be different, but the number of molecules in the meter stick doesn't change!

   (b) No: acceleration is measured in units of time, which changes from frame to frame.

   (c) No: due to time dilation different frames will record a different number of rotations per second.

   (d) Yes: the two events are both on your worldline and thus have a definitive causal ordering.

3. Relative to you, the γ factors associated with A and B are $5/3$ and $5/4$, respectively. So their lengths in your frame are $3L/5$ and $4L/5$, respectively. While overtaking B, A has to travel farther than B by an amount equal to the sum of the lengths of the trains: i.e. it must travel $7L/5$. The difference in speeds of the two trains as measured by you is $1/5$, so:

$$t_{\text{you}} = \frac{7L/5}{1/5} = 7L$$

4. This question is meant to be done with a ruler and pencil on paper or a graphing software and should involve as few equations as possible. The purpose is to study special relativity like velocity addition directly from measurements made on a spacetime diagram without much calculation.

(a) Setting the coordinates in Charlie's frame, we can write down the worldline of Bob and Alice

$$\text{Bob's time axis } ct_B\text{: } ct = \frac{c}{v_C^B}x = 2x, \tag{2.1}$$

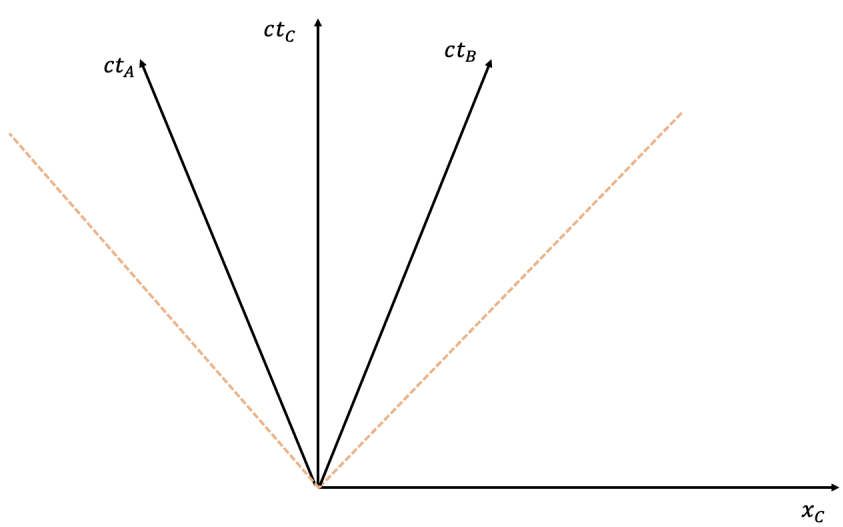
where Bob's speed in Charlie's frame is $v_C^B = c/2$, and

$$\text{Alice's time axis } ct_A\text{: } ct = \frac{c}{v_C^A}x = -2x, \tag{2.2}$$

where Alice's speed in Charlie's frame is $v_C^A = -c/2$.

Alice, Charlie and Bob meet when $t_C = 0$ according to Charlie (also $t_C = t_B = t_A = 0$), so they meet at the origin. The lightcone makes 45 degree angle with the Charlie's axes (slope equal to 1) while the worldline of Bob and Alice have slope 2 and -2 respectively, which is made evident by (2.1) and (2.2). See figure 2 for an illustration.

Figure 2:



(b) As we have shown in the previous problem, Alice's and Bob's lines of simultaneity are parallel to their own x axis, which is symmetric to their own time axis around the lightcone in the positive x direction. Therefore, we can directly add the x axis of Alice and Bob (denoted by x_A and x_B) on top of the previous figure (see figure 3).

Figure 3:

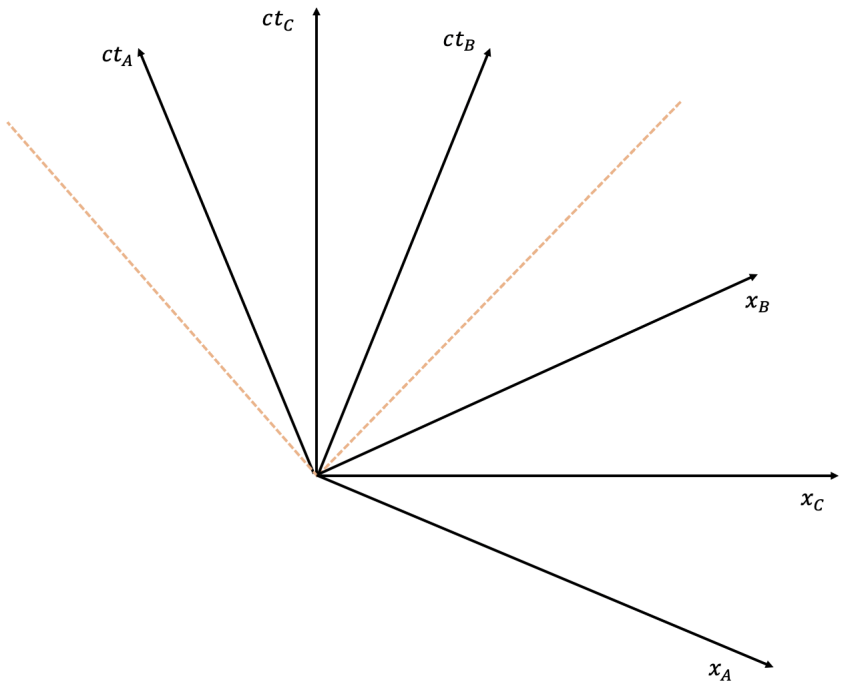
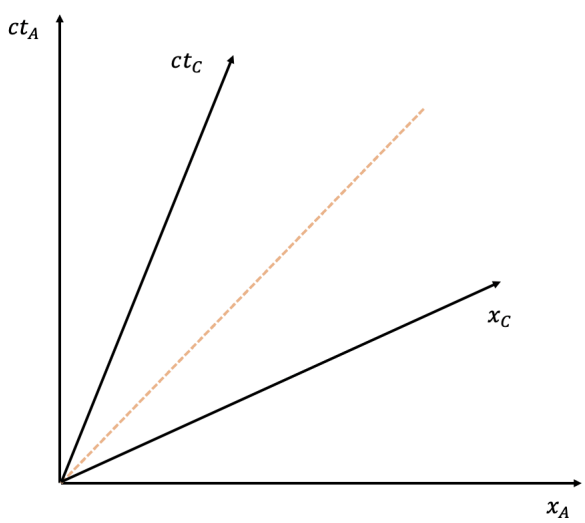


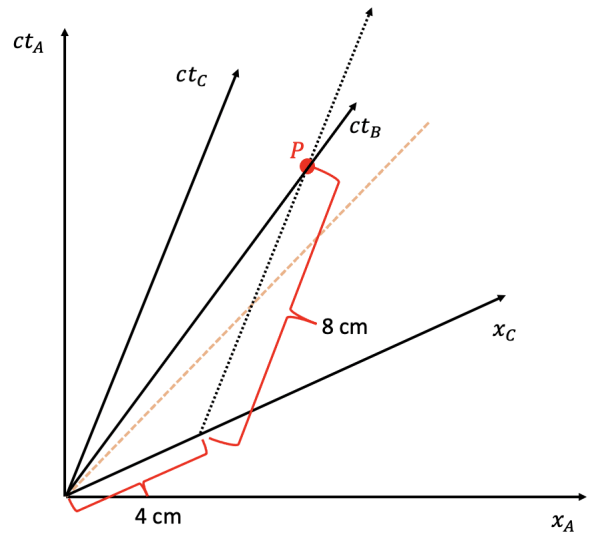
Figure 4:



(c) In Alice's frame, Charlie moves at speed $c/2$ on the right, so Alice sees Charlie the same way that Charlie sees Bob from figure 3. This is illustrated in figure 4.

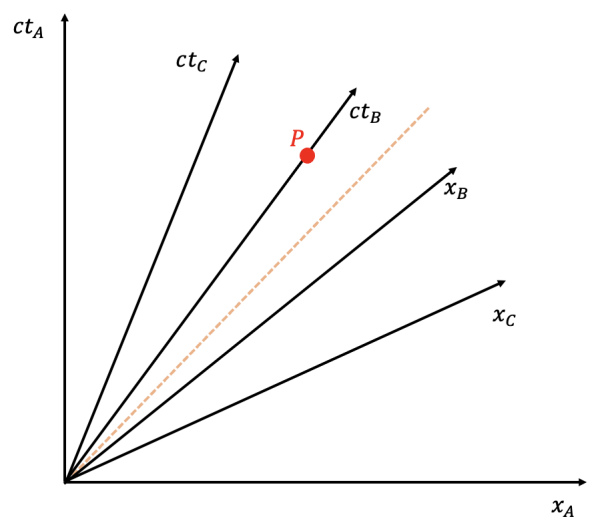
Now we want to add Bob's axes to figure 4. Note that Bob moves at speed $c/2$ on the right in Charlie's frame. We just need to find any point on the diagram whose coordinates satisfies $x_C = (c/2)t_C$, connect the origin with that point using a straight line, and that line will be the worldline (or equivalently the time axis) of Bob in Alice's frame. Figure 5 illustrates this process by choosing a point P whose coordinates are $x_C = 4$ cm and $ct_C = 8$ cm. Note that Charlie's positive time direction is parallel to the axis ct_C in Alice's frame.

Figure 5:



Now using the fact that the x axis of Bob should be symmetric to the time axis ct_B about the lightcone, we can directly add Bob's x axis (see figure 6).

Figure 6:

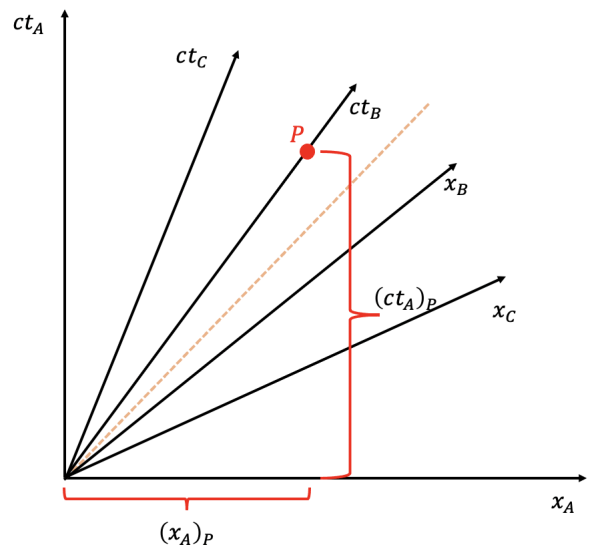


Now we can estimate Bob's speed in Alice's frame from figure 6 directly. All we need to do is to pick a point on Bob's time axis (which is also Bob's worldline), for example point P again, and then measure P's coordinates in Alice's coordinates $(ct_A)_P$ and $(x_A)_P$ (see figure 7). Then Bob's speed in Alice's frame (let's denote it by v_A^B) is given by:

$$\frac{v_A^B}{c} = \frac{(x_A)_P}{(ct_A)_P}. \tag{2.3}$$

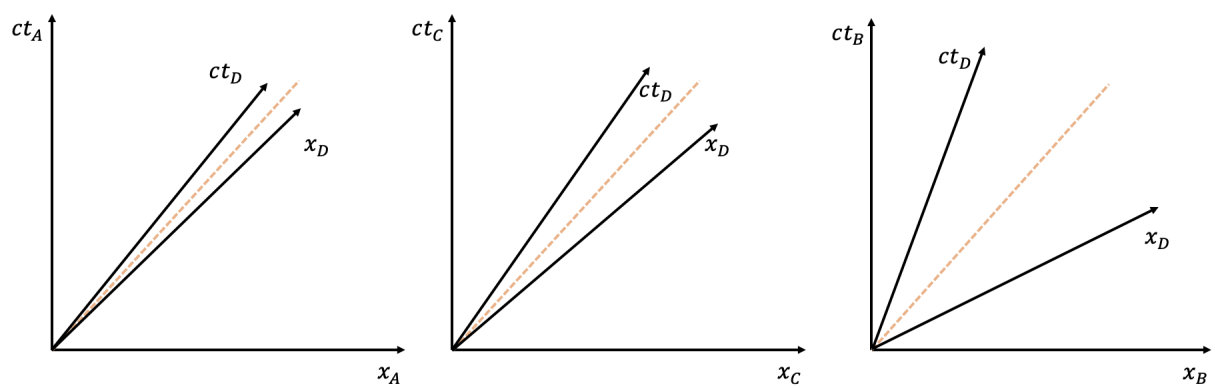
The result should be close to the value calculated using the velocity addition formula, which turns out to be $\frac{4}{5}c$.

Figure 7:



(d) Bob sees Delilah the same way that Charlie sees Bob, and Charlie sees Delilah the same way that Alice sees Bob. So, we already know what Delilah's axes look like in Bob's and Charlie's frames. Now, to find out Delilah's axes seen in Alice's frame, we can carry out a similar procedure from part (c), which we used to find Bob's axes in Alice's frame. We simple need to find a point with coordinates satisfying $x_B = (c/2)t_B$ and connect it to the origin, and that line will be Delilah's time axis seen in Alice's frame. The left, middle, and right panel of figure 8 shows Delilah's worldline and line of simultaneity in Alice's, Charlie's, and Bob's frame, respectively.

Figure 8:



Now we can use a similar procedure from part (c), which we used to find Bob's speed in Alice's frame to find Delilah's speed in Alice's frame, which should turn out to be close to $\frac{13}{14}c$, the result we get using the velocity addition formula. Apparently Delilah has the fastest speed according to Alice, and there is no frame where Delilah's speed is c.

5. (a) Knight #2 sees the starting line move towards him with speed v, so the distance is length contracted to ℓ_0/γ (where $\gamma = 1/\sqrt{1 - v^2/c^2}$).

   (b) It takes a time interval

$$\Delta t_2 = \frac{\ell_0}{\gamma v} \tag{1}$$

   for Knight #2 to cover the distance from their starting line to the finish line. Since the finish line is crossed at time zero, the start line is crossed at

$$t_2(E_2) = -\frac{\ell_0}{\gamma v} \ . \tag{2}$$

   (c) Knight #1's starting line approaches Knight #2 with velocity $\vec{v} = v\hat{x}$

   (d) Knight #1 is seen by Knight #2 to move in the x direction with speed

$$v_{1\,\text{according to}\,2} \equiv v_2 = \frac{2v}{1 + v^2/c^2} \ . \tag{3}$$

   (e) Remember, this is all as seen by Knight #2. This knight sees the starting line approach with speed v, and they see Knight #1 moving with speed v_2. They say that Knight #1 runs for a total time Δt defined as follows:

$$\begin{aligned} v_2 \Delta t &= \text{Total distance that the Knight \#1 moves after crossing their starting line} \\ &= \text{Distance from starting line to finish line according to Knight \#2+} \\ &\qquad \text{Distance that these lines move according to Knight \#2} \\ &= \ell_0/\gamma + v\Delta t \end{aligned} \tag{4}$$

   Solve this for Δt:

$$\Delta t = \frac{\ell_0/\gamma}{v_2 - v} \ . \tag{5}$$

   Let's look at $v_2 - v$:

$$\begin{aligned} v_2 - v &= \frac{2v}{1 + v^2/c^2} - v \\ &= \frac{2v - v(1 + v^2/c^2)}{1 + v^2/c^2} \\ &= \frac{v(1 - v^2/c^2)}{1 + v^2/c^2} \\ &= \frac{v}{\gamma(1 + v^2/c^2} \ . \end{aligned} \tag{6}$$

Using this, we find that the total time Knight #1 runs is

$$\Delta t = \frac{\gamma \ell_0 (1 + v^2/c^2)}{v} \ . \tag{7}$$

Since Knight #1 crosses the finish line at $t_2 = 0$, they must have started at

$$t_2(E_1) = -\frac{\gamma \ell_0 (1 + v^2/c^2)}{v} \ . \tag{8}$$

(f) Compare the time at which Knight #1 started with the time at which Knight #2 started: According to Knight #2, Knight #1 got a head start, beginning the race early by a time interval

$$\begin{aligned}
\delta t_2 &= t_2(E_2) - t_2(E_1) \\
&= -\frac{\ell_0}{\gamma v} + \frac{\ell_0 \gamma (1 + v^2/c^2)}{v} \\
&= -\frac{\ell_0 \gamma}{\gamma^2 v} + \frac{\ell_0 \gamma (1 + v^2/c^2)}{v} \\
&= -\frac{\ell_0 \gamma (1 - v^2/c^2)}{v} + \frac{\ell_0 \gamma (1 + v^2/c^2)}{v} \\
&= 2\frac{\ell_0 \gamma v}{c^2} \ .
\end{aligned} \tag{9}$$

Knight #2 thinks Knight #1 cheated by starting early!

(g) Repeating all this from the other point of view, we'll see that everything is symmetric: Knight #1 will believe that Knight #2 cheating by starting early. Neither of the two knights think this is a fair race.

(h)

$$(t_K, x_K)_{E_f} = (0, 0) \tag{10}$$

(i)

$$\begin{aligned}
t_2 &= \gamma (t_K + v x_K/c^2) = 0 \\
x_2 &= \gamma (x_K + v t_K) = 0
\end{aligned} \tag{11}$$

(j)

$$(t_K, x_K)_{E_2} = (-\ell_0/v, \ell_0) \tag{12}$$

(k)

$$t_2 = \gamma(t_K + vx_k/c^2) = \gamma(-\ell_0/v + v\ell_0/c^2) = \frac{\gamma\ell_0}{v}(-1 + v^2/c^2)$$
$$= -\frac{\ell_0}{v\gamma} \tag{13}$$
$$x_2 = \gamma(x_K + vt_k) = \gamma(\ell_0 - v\ell_0/v)$$
$$= 0 \tag{14}$$

(l)

$$(t_K, x_K)_{E_1} = (-\ell_0/v, -\ell_0) \tag{15}$$

(m)

$$t_2 = \gamma(t_K + vx_k/c^2) = \gamma(-\ell_0/v - v\ell_0/c^2)$$
$$= -\frac{\gamma\ell_0(1 + v^2/c^2)}{v} \tag{16}$$
$$x_2 = \gamma(x_K + vt_k) = \gamma(-\ell_0 - v\ell_0/v)$$
$$= -2\gamma\ell_0 \tag{17}$$